

A Machine Learning Model for the Prediction of Hessian Matrices based on Semiempirical Electronic Structure Theory

Machine learning (ML) models have become important tools in quantum chemistry.[1] For example, high dimensional neural networks and Gaussian process regression models were quite successful for the prediction of potential energy surfaces (PES) based on geometrical quantities.[2, 3]

In this work, we want to explore the applicability of ML for the prediction of the nuclear Hessian matrix. The Hessian matrix is used for the computation of harmonic nuclear vibrational frequencies, which are required, e.g., for the simulation of infrared spectra and the zero point vibrational energy. The target of the presented ML model are Hessian matrix elements. The Hessian matrix elements in the cartesian coordinate system are dependent on the orientation of the coordinate system. To achieve a unique mapping, we present a method to generate atomistic features in an invariant coordinate system, thus providing translational and rotational invariance. The model is free from element-specific parameters.

Here, we show the performance of the model in the prediction of vibrational frequencies for small molecules and benchmark it for the computation of zero point vibrational energies for large subsets from the GDB7 molecule data set.

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